Scrum Master: Nathanael Wolfe

Project:
ETSU Marketplace is a university-only online marketplace that allows verified students to buy, sell, and exchange items and housing opportunities securely. It is a marketplace designed specifically for students to interact with other verified students. The system is built using ASP.NET MVC and C# and focuses on creating a safe environment for students to conduct marketplace transactions.

Activities

Sprint 4 focused on improving the listing system by expanding the platform's image handling capabilities and improving search and filtering functionality. During this sprint, the team continued to meet during scheduled class time and used Discord to coordinate work outside of class. Sprint planning involved reviewing the Product Backlog, selecting tasks related to listing improvements, image handling, and filtering features, and moving those items into the Sprint Backlog.

Much of the development work during this sprint involved implementing image support for listings. This included creating database tables to store image references, implementing services for image upload and deletion, and updating controllers and models to support image management. Additionally, the team implemented improvements to search and filtering functionality so that users can narrow listings by categories more efficiently. These improvements significantly enhanced the usability of the marketplace interface and allowed listings to display images in both search results and listing detail pages.

The sprint also included continued testing of previously implemented features to ensure that listings, filtering, and image uploads work correctly together within the application.

Challenges

One of the main challenges during Sprint 4 was implementing image uploads and ensuring they integrate correctly with the listing system. Handling file uploads required careful backend implementation, including storing references to uploaded images in the database while ensuring that images are properly displayed on the frontend.

Another challenge involved updating multiple parts of the application simultaneously. Because image functionality touches several layers of the system—including models,

controllers, services, and views—changes needed to be coordinated carefully to avoid breaking existing features.

Additionally, implementing filtering functionality required adjustments to search queries and controller logic so that categories could be used to narrow down listing results efficiently.

Despite these challenges, the team was able to successfully implement the planned features and continue improving the marketplace system.

Achievements

By the end of Sprint 4, the ETSU Marketplace application gained several important improvements and new capabilities:

- Listings can now display images in search results and listing detail pages
- A database structure was created to store listing image references
- Image upload and deletion services were implemented
- Listings can now be filtered by category to improve browsing efficiency
- Listing pages now support multiple images for better item presentation

These enhancements improved the visual experience of the marketplace and made listings more informative and engaging for users.

Member Contributions & Owned Tasks

Jacob Sullins

User Stories Completed:

- Add multi-image upload functionality for listings
- Create services to handle image uploads and deletion

Work Completed:

- Implemented image upload service
- Integrated image handling with listing controllers
- Implemented backend logic for storing image references
- Assisted with database integration for listing images

Estimated Time Spent:

Task	Estimated Time
Designing image service	3 hours
Implementing upload functionality	4 hours
Database integration	3 hours
Testing and debugging	2 hours
Total Estimated Time	**12 hours**

Jabari Mitchell

User Stories Completed:

• Display listing images in search results and listing pages

Work Completed:

• Updated views to render listing images
• Integrated image display logic with Razor views
• Adjusted UI layout for listings with images
• Tested listing image display across different pages

Estimated Time Spent:

Task	Estimated Time
Frontend UI updates	3 hours
View integration	3 hours
Layout adjustments	2 hours
Testing and debugging	2 hours
Total Estimated Time	**10 hours**

Nathanael Wolfe

User Stories Completed:

• Implement category filtering for listings
• Improve listing search functionality

Work Completed:

• Implemented category filtering logic in controllers
• Updated search queries to support filtering
• Integrated filtering with listing views
• Tested filtering functionality with multiple listing types

Estimated Time Spent:

Task	Estimated Time
Designing filtering logic	3 hours
Controller implementation	3 hours
View integration	2 hours
Testing and debugging	2 hours
Total Estimated Time	**10 hours**

Palwasha Newell

User Stories Completed:

• Improve search functionality for listings

Work Completed:

• Updated search functionality to support category-based queries
• Adjusted controller logic for search filtering
• Updated views to reflect filtered search results
• Tested search functionality with multiple scenarios

Estimated Time Spent:

Task	Estimated Time
Search logic updates	3 hours
Controller updates	2 hours

Task	Estimated Time
View updates	2 hours
Testing and debugging	2 hours
Total Estimated Time	**9 hours**

Team Velocity

For Sprint 4, the team completed:

• 5 major development tasks
• Several backend and frontend integration improvements
• Full implementation of listing image functionality and filtering

Estimated total team development time:
41 combined hours

Average per team member:
10–11 hours

Compared to earlier sprints, Sprint 4 focused more on feature refinement and improving user experience rather than building foundational functionality. These improvements significantly enhanced the usability and visual presentation of the ETSU Marketplace platform.

Backlog Export

An exported copy of the Trello board backlog has been attached to this report. The backlog includes:

• Product backlog items
• Sprint backlog tasks
• Tasks currently in progress
• Completed tasks grouped by sprint

This export provides a complete overview of project progress and task status throughout the development cycle.

Backlog Export (Trello Board)

The following table summarizes the complete project backlog from the ETSU Marketplace Trello board. It includes items currently in the Product Backlog, tasks selected for Sprint 4, tasks in progress, and tasks completed in previous sprints.

Product Backlog

Task / User Story	Status	Sprint
Filter lease takeover requests by category	Product Backlog	Not started
View seller profile to see other listings	Product Backlog	Not started
Selenium Tests for SignalR Chats	Product Backlog	Not started
Integrate Selenium into GitHub Actions	Product Backlog	Not started
Show multiple images in listing details	Product Backlog	Not started

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Sprint 4 Backlog

Task / User Story	Status	Sprint
Search for listings	Done	Sprint 4
Filter listings by category	Done	Sprint 4
Add multi-image upload for listings	Done	Sprint 4
Show listing images in index/search pages	Done	Sprint 4
Show single image in listing details	Done	Sprint 4
Create service for image uploads/deletes	Done	Sprint 4
Modify controllers to support image uploads	Done	Sprint 4
Add migration for image database table	Done	Sprint 4

Task / User Story	Status	Sprint
Update models/views to support images	Done	Sprint 4
Add dropdown filters in HTML forms	Done	Sprint 4

Sprint 3 Completed Tasks

Task / User Story	Status	Sprint
Create API endpoint tests	Done	Sprint 3
Selenium tests for login	Done	Sprint 3
Selenium tests for register	Done	Sprint 3
Selenium tests for listing creation	Done	Sprint 3
Selenium tests for listing update	Done	Sprint 3
Selenium tests for listing deletion	Done	Sprint 3
Selenium tests for listing read	Done	Sprint 3

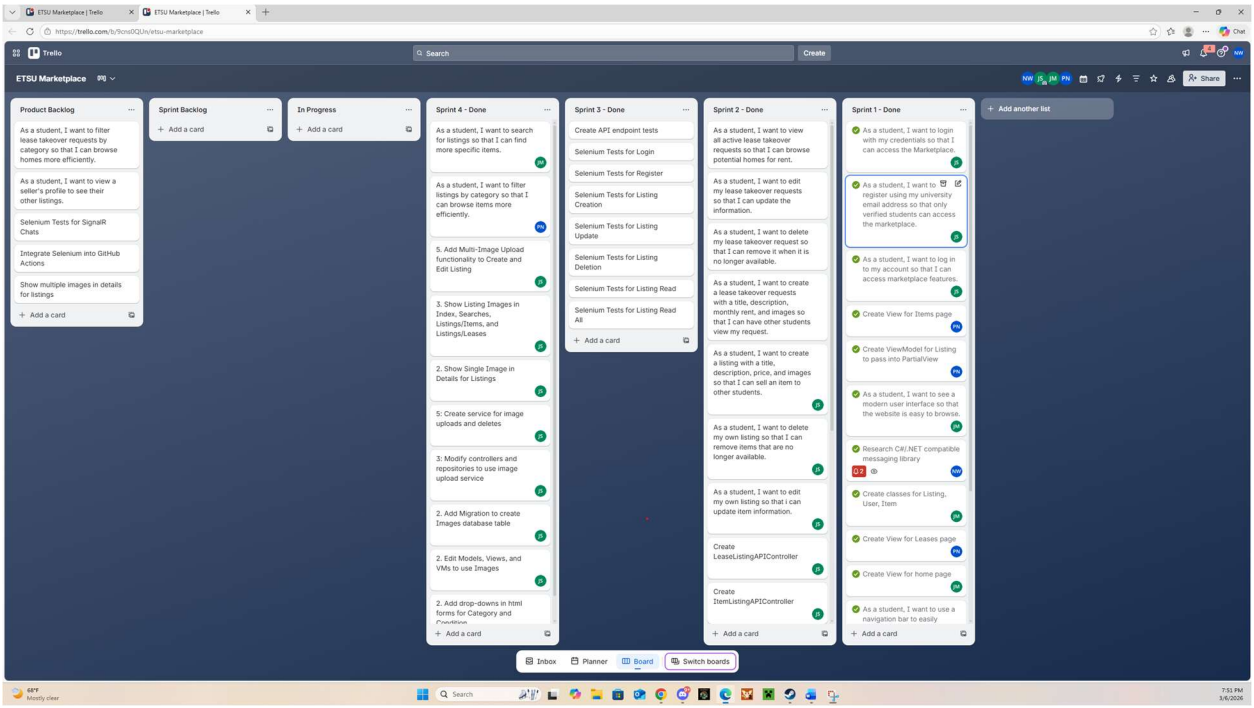
Sprint 2 Completed Tasks

Task / User Story	Status	Sprint
View active lease takeover requests	Done	Sprint 2
Edit lease takeover requests	Done	Sprint 2
Delete lease takeover requests	Done	Sprint 2
Create lease takeover requests	Done	Sprint 2
Create item listing	Done	Sprint 2
Delete item listings	Done	Sprint 2
Edit item listings	Done	Sprint 2

Task / User Story	Status	Sprint
Create LeaseListingAPIController	Done	Sprint 2
Create ItemListingAPIController	Done	Sprint 2

Sprint 1 Completed Tasks

Task / User Story	Status	Sprint
Login with credentials	Done	Sprint 1
Register with university email	Done	Sprint 1
User login functionality	Done	Sprint 1
Create Items page view	Done	Sprint 1
Create Listing ViewModel	Done	Sprint 1
Modern UI layout	Done	Sprint 1
Research messaging library	Done	Sprint 1
Create Listing/User classes	Done	Sprint 1
Create Lease view	Done	Sprint 1
Create home page view	Done	Sprint 1
Navigation bar for site	Done	Sprint 1


ETSU Marketplace | Trello
https://trello.com/b/9cns0QUn/etsu-marketplace
Trello
Search
Create
ETSU Marketplace
Share
Product Backlog
As a student, I want to filter lease takeover requests by category so that I can browse homes more efficiently.
As a student, I want to view a seller's profile to see their other listings.
Selenium Tests for SignalR Chats
Integrate Selenium into GitHub Actions
Show multiple images in details for listings
Add a card
Sprint Backlog
Add a card
In Progress
Add a card
Sprint 4 - Done
As a student, I want to search for listings so that I can find more specific items.
As a student, I want to filter listings by category so that I can browse items more efficiently.
5. Add Multi-Image Upload functionality to Create and Edit Listing
3. Show Listing Images in Index, Searches, Listings/Items, and Listings/Leases
2. Show Single Image in Details for Listings
5: Create service for image uploads and deletes
3: Modify controllers and repositories to use image upload service
2. Add Migration to create Images database table
2. Edit Models, Views, and VMs to use Images
2. Add drop-downs in html forms for Category and Condition
Add a card
Sprint 3 - Done
Create API endpoint tests
Selenium Tests for Login
Selenium Tests for Register
Selenium Tests for Listing Creation
Selenium Tests for Listing Update
Selenium Tests for Listing Deletion
Selenium Tests for Listing Read
Selenium Tests for Listing Read All
Add a card
Sprint 2 - Done
As a student, I want to view all active lease takeover requests so that I can browse potential homes for rent.
As a student, I want to edit my lease takeover requests so that I can update the information.
As a student, I want to delete my lease takeover request so that I can remove it when it is no longer available.
As a student, I want to create a lease takeover requests with a title, description, monthly rent, and images so that I can have other students view my request.
As a student, I want to create a listing with a title, description, price, and images so that I can sell an item to other students.
As a student, I want to delete my own listing so that I can remove items that are no longer available.
As a student, I want to edit my own listing so that i can update item information.
Create LeaseListingAPIController
Create ItemListingAPIController
Add a card
Sprint 1 - Done
As a student, I want to login with my credentials so that I can access the Marketplace.
As a student, I want to register using my university email address so that only verified students can access the marketplace.
As a student, I want to log in to my account so that I can access marketplace features.
Create View for Items page
Create ViewModel for Listing to pass into PartialView
As a student, I want to see a modern user interface so that the website is easy to browse.
Research C#/.NET compatible messaging library
Create classes for Listing, User, Item
Create View for Leases page
Create View for home page
As a student, I want to use a navigation bar to easily
Add a card
Add another list
Inbox
Planner
Board
Switch boards
68°F
Mostly clear
Search
7:51 PM
3/6/2026